SCIDA: Self-Correction Integrated Domain Adaptation From Single- to Multi-Label Aerial Images

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***Abstract*— Most publicly available datasets for image classification are with single labels, while images are inherently multi-labeled in our daily life. Such an annotation gap makes many pretrained single-label classification models fail in practical scenarios. For aerial images, this annotation issue is more concerned: Aerial data naturally cover a relatively large land area with multiple labels, while annotated aerial datasets currently publicly available (e.g., UCM and AID) are single-labeled. As manually annotating multilabel aerial images (MAIs) would be time-/labor-consuming, we propose a novel self-correction integrated domain adaptation (SCIDA) method for automatic multilabel learning. SCIDA is weakly supervised, i.e., automatically learning the multilabel image classification model from using massive, publicly available single-label images. To achieve this goal, we propose a novel labelwise self-correction (LWC) module to better explore underlying label correlations. This module also makes the unsupervised domain adaptation (UDA) from single-label to multilabel data possible. For model training, the proposed method uses single-label information yet requires no prior knowledge of multilabeled data and predicts labels for MAIs. Through extensive evaluations, the proposed model, which is trained with single-labeled MAI-AID-s and MAI-UCM-s datasets, achieves much better performances than comparative methods on our collected multiscene aerial image dataset. The code and data are available on GitHub (https://github.com/Ryan315/Single2multi-DA).**

***Index Terms*— Aerial image, graph convolutional network (GCN), MAI dataset, noise, OpenStreetMap (OSM), unsupervised domain adaptation (UDA).**

I. Introduction

With easy access to increasing aerial data from satellites/unmanned aerial vehicles (UAVs), annotating the newly collected aerial data is of great importance. However, obtaining clean multilabel annotations manually for aerial data has long been a challenging task. A recent trend for aerial data annotation is resorting to crowdsourcing data, such as OpenStreetMap (OSM). OSM, an editable map, is built/annotated by volunteers from scratch. However, the quality of such a manually annotated map may not be satisfying. Incompleteness and incorrectness are two primary concerns, as shown in the example in Fig. 1.

Instead of resorting to crowdsourcing data, recent advances in machine learning (ML) make automatic annotation of aerial images possible. To train a reliable multilabel aerial image (MAI) classification framework, we need to design an efficient ML model architecture and learn with massive annotated data. However, collecting such MAIs with an exhaustive, consistent list of annotations requires significant time and effort. Furthermore, there is almost no publicly available MAI dataset, and the online annotations from OSM are not always reliable. Training the model with noisy labels from OSM could lead to poor classification performance. Therefore, direct training of a MAI classification model remains challenging.

An alternative way is to train the model with single-labeled aerial image data since annotating a single-label aerial image is much easier than a multilabel one. Moreover, there are publicly available single-labeled datasets, e.g., AID and UCM datasets, which have covered almost all classes of interest for aerial images. Therefore, they could provide intense supervision for training a multilabel classification model. However, using the model trained by single-label data to predict the labels for a multilabel image is a challenging task. This task is shown in Fig. 1, where single-labeled data (the source domain) are shown in Fig. 1(a) and the corresponding multilabel data including the same labels (the target domain) is shown in Fig. 1(b). To realize MAI classification using single-labeled data, we formulate the problem as a domain adaptation (DA) task and propose a novel framework where prior information from single-label data can be adapted to MAIs.

To our knowledge, this is the first work to tackle the challenging task of learning an MAI classifier on large-scale dataset (the target domain) by using publicly available single-label data (the source domain). Our major contributions are given as follows.

1) We propose a challenging single-label-to-multilabel DA task. The target domain multilabel data are

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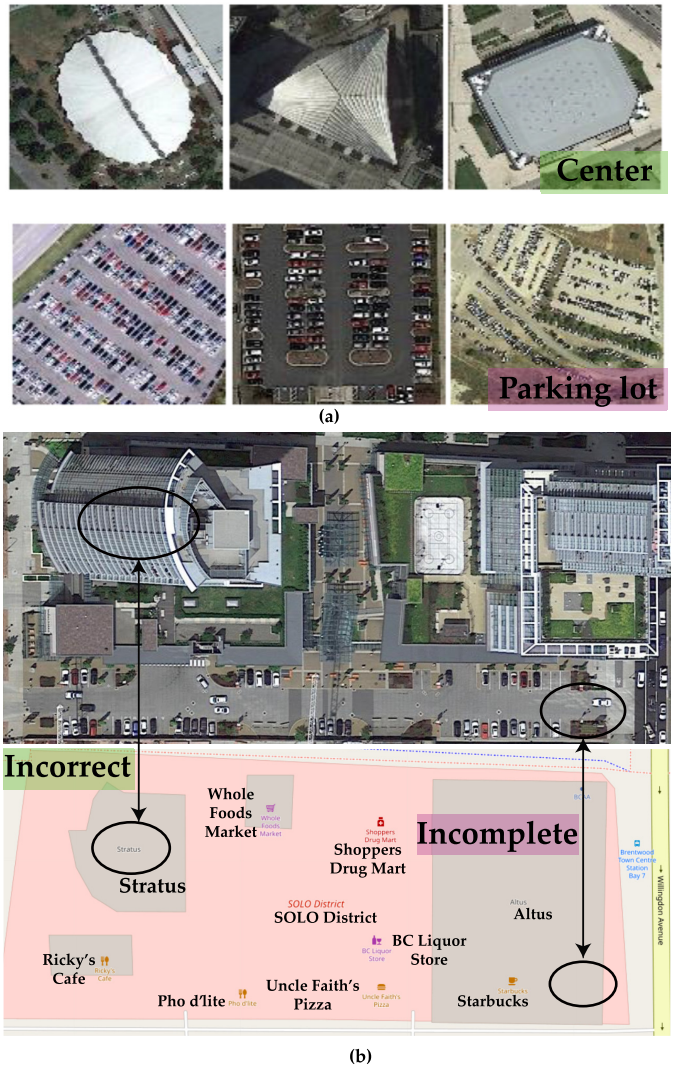


Fig. 1. (a) Single-label aerial image examples from the MAI-AID-s dataset, which serves as the source domain data. (b) (Top) MAI from the MAI dataset, which serves as the target domain data and (Bottom) corresponding noisy annotations from OSM for the top image. As indicated, "Center" is incorrectly annotated as "Stratus" and "Parking lot" is missed.

unannotated, large-scale, unconstrained aerial images in real-world scenarios, while the source domain is annotated single-label aerial data publicly available (e.g., AID [1] and UCM [2]).

2) We propose a novel self-correction integrated domain adaptation (SCIDA) framework, from single-label to MAIs. Different from existing feature-based unsupervised domain adaptation (UDA) methods, our framework explores the underlying label correlations by introducing the labelwise self-correction (LWC) module. With the graph convolutional network (GCN) backbone, the LWC explores label correlations and iteratively corrects pseudo labels learned by our DA module (the domainwise correlation (DWC) branch in Fig. 2).
3) We compare labeling strategies for multilabel datasets, to study the learning potential of using single labels. Given the same annotation budget, our experiments show that the networks trained with single-label images yield competitive performances as those learned with a fully annotated subset of multilabel images.
4) A new single-MAI dataset with clean labels is collected in our study to make the experiment possible. It is the first web available dataset for single- MAIs.

II. Related Work

A. Partial Multilabel Learning

Multilabel learning has been an active research topic of practical importance, as images collected in the wild are always with more than one annotation [3], [4]. Conventional multilabel learning research [5]–[9] mainly relies on the assumption that a small subset of images with full labels are available for training. However, this may be difficult to satisfy in practice, as manually yielded annotations always suffer from incomplete and incorrect annotation problems, as shown in Fig. 1. Therefore, partial multilabel learning (PML) is emerging, which aims to learn a multilabel classification model from ambiguous data [10]–[12]. Traditional methods treat missing labels as negatives and wrong labels as positives during the model training process [13]–[17], while it could lead to degradation of the classification performance.

To mitigate the problem of missing or wrong labels, a novel approach for partial label learning (PLL) is to treat missing labels as a hidden variable via probabilistic models and predict missing labels by posterior inference [18], [19]. The work in [20] models missing labels as negatives and corrects the induced error by learning a transformation on the output of the multilabel classifier. However, this approach requires high memory and is hard to optimize. Scaling these models to large datasets would be difficult [21], [22]. Another recent trend for PLL can be found in [23]–[25], which introduces curriculum learning and bootstrapping to increase the number of annotations. During model training, this approach uses partially annotated data and unannotated data whose predicted labels are with the highest confidences [26], [27]. Curriculum learning [28] is further combined with the graph model in [29] to better capture the label association when exploiting the unlabeled data during training. This approach still relies heavily on unlabeled images, which would taint the training data when they are attached to incorrect labels. This problem is called semantic drift.

Different from existing models, we propose a novel approach for multilabel learning with the extreme case of partial multilabel data, i.e., the single label data. Compared with commonly used partial multilabel data, single-label data are collected specifically for a single class. Such data are much easier to acquire online and do not have the incomplete or incorrect annotation problem. In this article, for the first time, we introduce DA to train the multilabel classification model using the annotated single-label data.

B. Domain Adaptation

It is a method to share knowledge between data from different datasets. DA aims to minimize the data gap between datasets [30]–[32]. Here, we formulate the knowledge transfer

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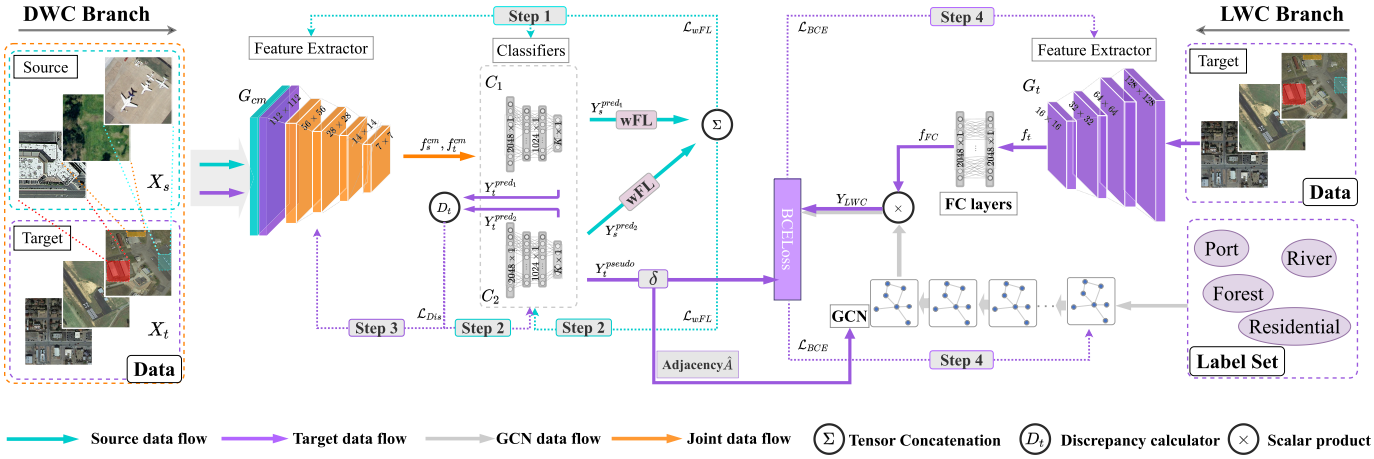


Fig. 2. Flowchart of training the proposed SCIDA with the ResNet backbone. The flow mainly consists of the DWC branch and the LWC branch. G_{cm} and G_t are feature generators and C_1 and C_2 represent classifiers. "wFL" means "weighted focal loss" and δ controls the learning depth at each training step.

between the single-label data and multilabel data as a DA problem in our task. Recent years have witnessed the exploitation of adversarial DA, which stems from the technique proposed in [33]. The principal idea is to introduce adversarial learning by one feature generator and one domain discriminator [34]–[36]. The generated features from the source domain and the target domain are aligned to confuse the domain discriminator until it cannot figure out which domain the features are from [37]–[40]. One major limitation of existing adversarial DA methods is that they are not task-specific. For instance, the generated features in the DA model might not work well for the classifier [41]–[43]. This problem could be even more severe for our task since the classification tasks for the source and target domains are two different types (one is single-label classification and the other is multilabel classification). A recent advance for task-specific DA is the maximum classifier discrepancy (MCD) method, which is proposed to make the adversarial mechanism task-specific by constructing adversarial learning between task-specific classifiers and feature generator [44]–[46]. However, this method could not generalize well when the classifiers from two domains are not the same. To solve this problem, here, we propose our new DA framework.

III. Method

In this section, we present the proposed SCIDA model for single-label-to-multilabel DA.

A. Overview

In the proposed SCIDA framework, we need to correlate the domainwise data and also explore the labelwise correlation among target domain data. This labelwise correlation is used for self-correction in the framework.

For the DWC, we propose using DA to correlate the two domains, due to the large domain gap between single-label and multilabel data. For the LWC, due to the lack of correlation for the one-hot encoded source domain data, the LWC is learned with self-supervision in the target domain. The GCN is introduced to model the LWC directly. The LWC is used for self-correction for multilabel classification.

The general flowchart of the proposed model is shown in Fig. 2, which mainly includes the DWC branch and the LWC branch. The inputs of the proposed framework are introduced as follows. The annotated source domain data are represented with $X_s = \{x_s^i, y_s^i\}_{i=1}^{N_s}$ (x and y represent the data and the label, respectively), while the target domain data are represented with $X_t = \{x_t^i\}_{i=1}^{N_t}$, where N_t represents the number of images in the dataset.

B. Domainwise Correlation

For the DWC, the goal of this branch is to align the features from source and target domains by utilizing two task-specific classifiers as the discriminator. The output of this branch is the pseudo labels for target domain data. This branch is made up of three parts: a common feature extractor G_{cm}, two classifiers C_1 and C_2, and a classifier discriminator D_t. The extracted source domain feature $f_s^{\text{cm}} = G_{\text{cm}}(X_s; \theta_{G_{\text{cm}}})$ and the target domain feature $f_t^{\text{cm}} = G_{\text{cm}}(X_t; \theta_{G_{\text{cm}}})$ are the inputs to the two classifiers. We need to detect target samples that largely deviate from the distribution of the source data and align features from the two domains. As two classifiers are assumed to be effective on source domain samples with full annotations, the classification results from these two classifiers should be the same. While the target samples deviate from the source data distribution and are likely to be classified differently by the two distinct classifiers, our goal is to minimize the performance gap between the two classifiers for target domain samples. In our framework, this discrepancy for the two classifiers can be calculated in the target domain by D_t. If the discrepancy is minimized, we assume that the data feature from the target domain is aligned with the source domain. Then, we can use the source domain annotations to supervise the classification of the target domain data. The general training of this branch can be found in Section IV.

However, different from regular task-specific DA in [44], the major challenge for DA from single-label to multilabel

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is the sharing of classifiers. If the single-label classifier is trained to generate multiple labels by setting a threshold for probability, the single-label source data will suffer from the problem of imbalanced training data. The imbalance issue contains two aspects: the imbalance of positive and negative samples in each class, and the imbalance of the samples in different classes in the entire dataset. In this task, there are much fewer positive samples than the negative samples, with a ratio of about $1/K$ (K is the number of classes). To overcome this problem, we propose to use the weighted focal loss (wFL) instead of the regular cross-entropy loss for the optimization of the model, which is formulated as follows:

$$\mathcal{L} = -\sum_{i=1}^{K} p_\beta \Big(\alpha y^i \left(1 - p^i\left(y^i|x\right)\right)^\gamma \log p^i\left(y^i|x\right) + (1-\alpha)\left(1-y^i\right) p_i^\gamma \left(y^i|x\right) \log\left(1-p^i\left(y^i|x\right)\right)\Big) \tag{1}$$

where p_β is a factor of classwise samples with respect to all the data in the dataset and fulfills $p_\beta \in (0, 1)$ & $\sum_{\beta=1}^{C} p_\beta = 1$. α and γ here are empirically set as 0.25 and 2, respectively [47].

For the multilabel target dataset, the imbalance is eased to some extent. However, as annotations are not available for the target domain data, the classification in the target domain does not generate a loss for the supervision of the model. Instead, the discrepancy between the two classifiers in the target domain is regarded as the loss and used to optimize the model.

C. Labelwise Correlation

For the LWC branch, the goal of this branch is to self-correct the pseudo label generated by the DWC branch. This branch consists of two components, including a separate target convolution neural network with G_t being the backbone for image classification and a GCN being the backbone for label correlation learning.

1) Correlation Matrix Construction: Labelwise correlation works by propagating label information between nodes based on the correlation matrix. It is a crucial point of how to construct the correlation matrix. As in the proposed unsupervised scenario, there is no predefined correlation matrix in the target task. We will construct the correlation matrix in a target-data-driven approach based on the pseudo ground-truth labels. The procedure of constructing the correlation matrix is detailed in Algorithm 1. For the pseudo ground truth, we introduce a hyperparameter δ for controlling the number of pseudo ground-truth labels for each image, as shown in Fig. 3. As there is no prior label knowledge for the target domain data and the total number of labels for each image varies, we assume that the percentage of pseudo ground-truth labels per image is a fixed constant δ (e.g., suppose that the dataset has 20 labels in total, with $\delta = 0.2$, the number of pseudo ground-truth labels per image is 4). δ is used to determine how many nodes in GCN need to be updated in each iteration. An intuition is that δ is set as the average number of labels per image in the pseudo ground truth. Besides, as the LWC branch is supervised by the pseudo

Algorithm 1 Correlation Matrix Construction

Stage I: Calculate the co-occurrence label
Input: Pseudo ground truth Y_t^{pseudo}
Output: Correlation matrix A;
while *epoch* $\leq$ *max epoch* **do**
  for *batch* $\leftarrow 1$ **to** N **do**
    for *idx_i, idx_j* $\leftarrow 1$ **to** *num_categories* **do**
      if $Y_t^{pseudo}[idx_i]$ & $Y_t^{pseudo}[idx_j] \neq 0$ **then**
        cor_count[idx_i][idx_j] $+= 1$
      end
    end
  end
end
Stage II: Correlation matrix normalization
for *idx* $\leftarrow 1$ **to** *num_categories* **do**
  cor_count[idx] $= \frac{cor_count[idx]}{\sum cor_count[idx][:]}$
end
return cor_count

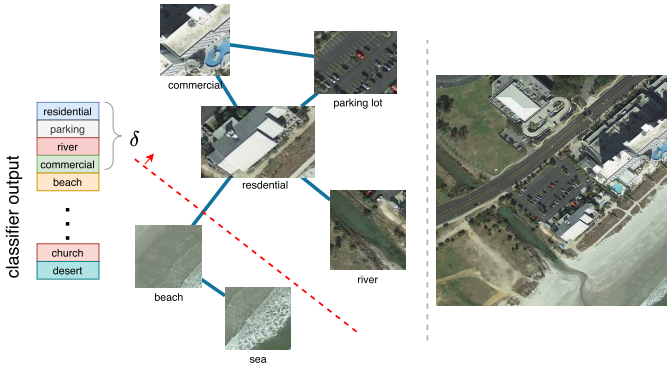


Fig. 3. Example of δ. (Right) image. (Left) Predicted labels.

labels generated by the DWC branch, introducing δ will help to smooth the decision boundary of the classification. In this way, the LWC could converge toward the correct direction and meanwhile is not restricted to the output of DWC branch. What is more, when calculating the co-occurrence of labels, some rare co-occurrence in the labels will introduce much noise and cause a long-tail distribution problem. Introducing the parameter δ could solve the long-tail distribution problem to a great extent. In the ablation study, we also analyzed the effects of the parameter δ.

2) Labelwise GCN: The inputs of this branch include four parts: the original target domain data, the target domain label set, the pseudo ground truth, as well as the normalized adjacency matrix $\hat{A}$ learned from the DWC branch (which represents the occurrence frequency of each label). We also need to point out that the label in the GCN module is different in a conventional convolution neural network and is intended to solve the problem under a non-Euclidean topological graph. The computation graph is generated based on the label embedding of each node and its neighbors. We follow a common

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Algorithm 2 Training for SCIDA

Input: X_s, X_t, Y_s, $\hat{A}$, labels;
Output: Parameters for G_{cm}, C_1, C_2, GCN, and G_t;
while *epoch* $\leq$ *max epoch* **do**
  for *batch* $\leftarrow 1$ **to** N **do**
    Pseudo labels generation: Input the normalized source and target domain data to optimize DWC branch by minimizing Eq. (3), (4), (5), and generate Y_t^{pseudo};
    Self Correction: Update both DWC and LWC by minimizing Eq. (6), which indicates the difference between Y_t^{pseudo} and Y_{LWC}.
  end
end

practice [48] to deploy GCN:

$$H^{(l+1)} = \sigma\left(\hat{A}H^{(l)}W^{(l)}\right) \tag{2}$$

where $\hat{A}$ is the normalized adjacency matrix mentioned above, $H^{(l)}$ denotes the label embedding at the lth layer in GCN, $W^{(l)}$ is a learnable transformation matrix, and σ acts as a nonlinear operation. We employ LeakyReLU to implement this operation.

The general operation routine for this branch is given as follows. The feature vector of target domain data is first extracted by $G_t(x_t; \theta_{G_t})$. This feature vector is fed to two fully connected (FC) layers and a 2048×1 feature vector f_{FC} is generated. For the input of the GCN model, GloVe [49] is used to generate the embedding of labels. Then, the output of GCN (a $2048 \times K$ matrix) and f_{FC} is fed to a scalar product layer to produce a classification result Y_{LWC} for this branch. The difference between the classification result $Y_t^{\mathrm{pred}_2}$ from C_2 (we assume that C_1 and C_2 can get a unified result finally) and Y_{LWC} is used to generate a binary cross-entropy loss (BCE loss), which will optimize all components in the LWC branch.

IV. Two-Stage Model Training

In the training procedure, we learn the parameters of DWC and LWC branches jointly and iteratively. In the primary stage, we initialize the pseudo label by the DWC branch. In the second stage, both branches are trained to optimize the pseudo label. The iterative training way is introduced as follows and concluded in Algorithm 2.

A. Pseudo Labels Generation in DWC

Pseudo labels generation is done by an adversarial DA in the DWC branch [44]. We first initialize the weights of the feature generator G_{cm} and the classifiers C_1 and C_2 and freeze other components. The model in this stage is optimized by the weighted focal loss calculated on the annotated source domain data. This process is formulated as

$$\min_{\theta_{G_{\mathrm{cm}}},\theta_{C_1},\theta_{C_2}} \mathcal{L}_{\mathrm{wFL}}(X_s, \mathbf{Y}_s) \tag{3}$$

where $\mathcal{L}$ is defined in (1).

We initialize the parameters of G_{cm}, C_1, and C_2 and train the framework for classification in the source domain, as well as to achieve an adversarial training ready state. Then, we use an adversarial manner to train the two classifiers C_1 and C_2 using the domain discriminator D_t and optimize the weights of G_{cm}.

To be specific, we first maximize the discrepancy loss generated by D_t for target domain data. This loss is used for optimizing C_1 and C_2. The purpose is to identify target samples whose extracted features deviate the most from the distribution of the source domain features. At the same time, we need to keep C_1 and C_2 effective for the classification of the source domain data. We formulate this by

$$\min_{\theta_{G_{\mathrm{cm}}},\theta_{C_1},\theta_{C_2}} \mathcal{L}_{\mathrm{wFL}}(X_s, Y_s) - \mathcal{L}_{\mathrm{dis}}\left(Y_t^{\mathrm{pred}_1}, Y_t^{\mathrm{pred}_2}\right) \tag{4}$$

where $Y_t^{\mathrm{pred}_1}$ and $Y_t^{\mathrm{pred}_2}$ represent multilabel predictions from C_1 and C_2, respectively, and $\mathcal{L}_{\mathrm{dis}}$ is computed with a scalar subtraction between these two [44].

Then, adversarial to (4), we minimize the classifier discrepancy loss by optimizing the weights of G_{cm}, in order to encourage uniformed classification results from the two classifiers. C_1 and C_2 are frozen now, and the objective function is

$$\min_{\theta_{G_{\mathrm{cm}}}} \mathcal{L}_{\mathrm{dis}}\left(Y_t^{\mathrm{pred}_1}, Y_t^{\mathrm{pred}_2}\right). \tag{5}$$

Finally, we assume Y_t^{pred} from C_2 to be Y_t^{pseudo}. We need to point out that Y_t^{pred} from C_1 is equal to Y_t^{pred} from C_2 (Y_t^{pseudo}) when the training converges.

B. Self-Correction

It is achieved unsupervised in the target domain for self-correcting the pseudo label Y_t^{pseudo}. Only by learning the label correlation with GCN (which can be understood as a self-supervision) but without extra supervisions/annotations, Y_t^{pseudo} is optimized and corrected.

In this stage, we first get Y_{LWC} from the LWC branch for the target domain data. This obtained Y_{LWC} from the scalar product operation (as in Fig. 2) is used for self-correction for the pseudo label Y_t^{pseudo}. Note that here, Y_{LWC} is the label predictions of target domain samples from the LWC branch. Y_{LWC} can be represented as $Y_{\mathrm{LWC}} = \{\hat{y}^1, \hat{y}^2, \ldots, \hat{y}^N\}$, for in total N target domain samples. As we note that the choice of different losses in this stage does not make a noticeable difference, we empirically choose the BCE loss

$$\min_{\theta_{G_t},\theta_{\mathrm{GCN}},\theta_{G_{\mathrm{cm}}},\theta_{C_1},\theta_{C_2}} \mathcal{L}_{\mathrm{BCE}}\left(Y_{\mathrm{LWC}}, Y_t^{\mathrm{pseudo}}\right) \tag{6}$$

and $\mathcal{L}_{\mathrm{BCE}}$ for sample x_i is further defined as

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{K}\sum_{i=1}^{K}\left(y^i \log\left(\hat{y}^i\right) + \left(1 - y^i\right)\log\left(1 - \left(\hat{y}^i\right)\right)\right) \tag{7}$$

where y^i is the pseudo labels from DWC branch. We need to point out that both y^i and $\hat{y}^i$ are variables being optimized, and the final convergence is only realized when y^i equals $\hat{y}^i$. This BCE loss will be used to optimize the parameters of the whole network, including both DWC and LWC branches.

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TABLE I
PROPERTIES OF THE PROPOSED MAI DATASET

Dataset	Size	# Images	# Categories	# Avg. Labels	# Max. Labels	# Min. Labels	Resolution
MAI-AID-s	1.7G	7,050	20	Single label annotation			600×600
MAI-AID-m	198.0M	3,239	20	3.73	9	1	512×512
MAI-UCM-s	176.3M	1,700	17	Single label annotation			256×256
MAI-UCM-m	105.5M	1,799	17	3.14	7	1	512×512

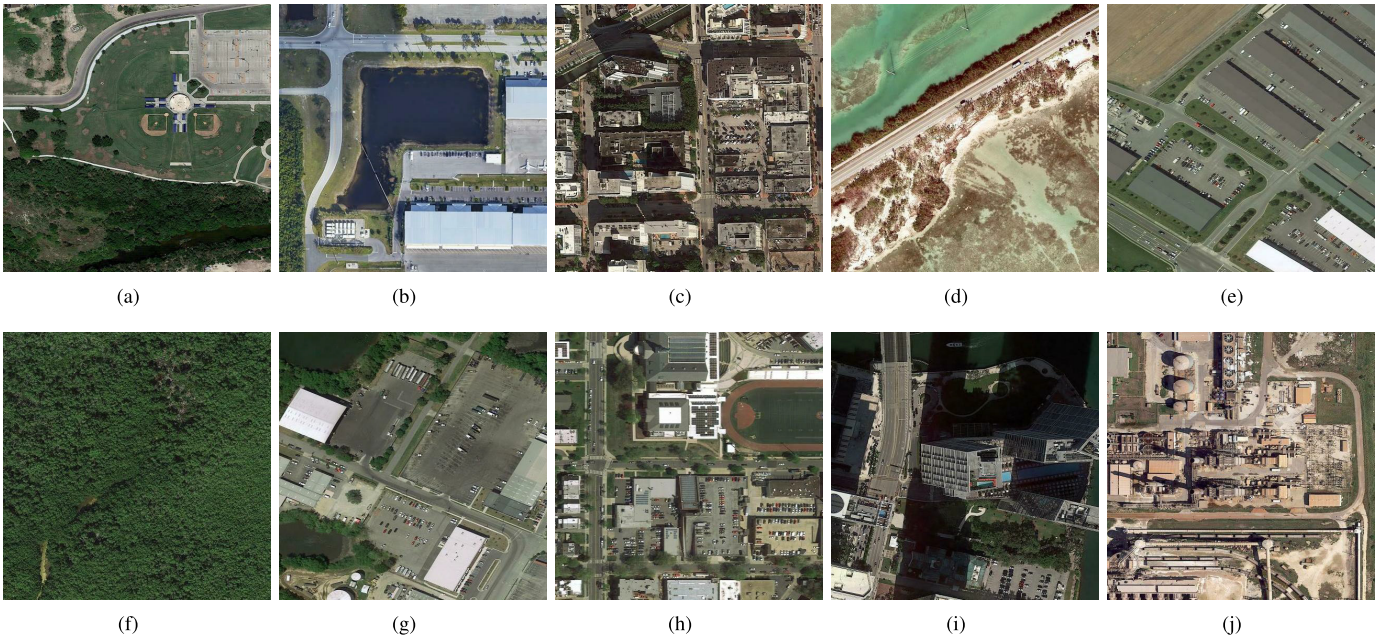


Fig. 4. Examples images with multiple labels from the MAI-AID-m dataset. (a) Baseball, parking lot, park, and river. (b) Airport, forest, and lake. (c) Commercial, bridge, parking lot, and residential. (d) Beach and bridge. (e) Farmland, commercial, and parking lot. (f) Forest and river. (g) Commercial, lake, parking lot, residential, and storage tank. (h) Stadium, soccer, residential, and parking lot. (i) River, bridge, and commercial. (j) Storage tank, train station, and works.

These two stages will iterate until convergence in the training process. Finally, Y_t^{pseudo} will stay unchanged, and we assume it as Y_{pred} in the testing phase.

V. EXPERIMENTS

A. Datasets, Setup, and Evaluation Metrics

1) MAI Dataset: Data are playing a critical role in enabling computers to interpret images as compositions of objects. Based on the scenario described above, we created a new dataset named MAI, which contains images and ground-truth annotations for the single-label-to-multilabel DA task. The proposed MAI dataset contains two subsets, MAI-AID and MAI-UCM. Besides, each subset is in pairs, which contains one single-label dataset and one multilabel dataset. The datasets are different in several ways and the statistical parameters of the datasets are given in Table I and Fig. 7.

a) MAI-AID: As a large-scale aerial image dataset, AID was introduced in 2017 [1], with sample images from Google Earth imagery. The original AID dataset has 30 categories, including 10 000 images; 7050 images from 20 categories are collected from the AID dataset and used as the single-label dataset named **MAI-AID-s** and 3239 images with exactly the same 20 categories are collected from OSM [50], which was a collaborative project to create a free editable geographic database of the world and named **MAI-AID-m**. The difference lies in the fact that the annotation of MAI-AID-s is single-label based, while the annotation of MAI-AID-m is multilabel-based. All images in MAI-AID-m are labeled by the specialists in remote sensing image interpretation. Some samples of the MAI-AID-m dataset are shown in Fig. 4.

b) MAI-UCM: UC Merced Land Use dataset (UCM) [2] is an aerial image dataset whose images are manually extracted from the USGS National Map Urban Area Imagery collection for various urban areas around the country. The original UCM dataset has 21 categories, including 2100 images; 1700 images from 17 categories are collected from the UCM dataset and used as the single-label dataset named **MAI-UCM-s** and 1799 images with exactly the same 17 categories are collected from OSM and named **MAI-UCM-m**. Similar to MAI-AID-m, the images of the MAI-UCM-m are also annotated with multilabels. Fig. 5 shows four examples of MAI-UCM-m dataset with object-level annotations.

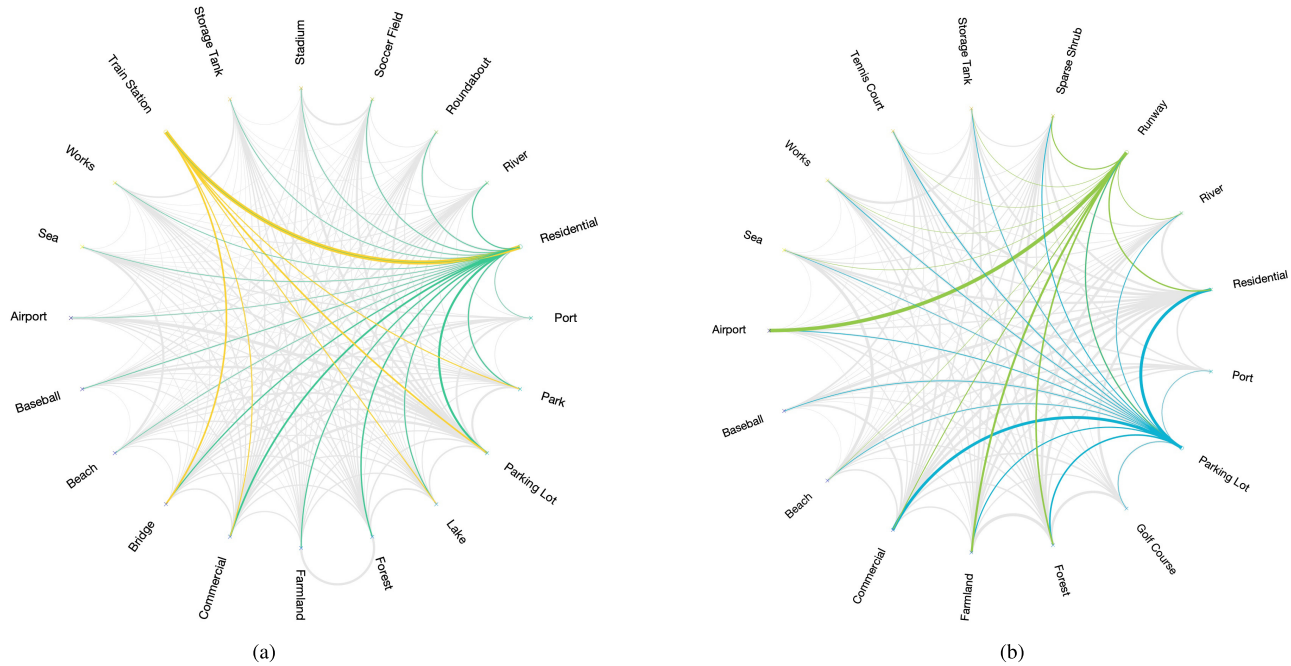
Meanwhile, for multilabel datasets, there exist inner connections and correlations between different categories. In Fig. 6, the adjacent matrix of the two multilabel datasets is visualized

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(a) (b) (c) (d)

Fig. 5. Examples with object-level annotations from the MAI-UCM-m dataset. Annotations are: (a) beach, sea, residential, forest, and parking lot; (b) airport, farmland, parking lot, and runway; (c) commercial, residential, forest, parking lot, baseball, and tennis; and (d) sparse shrub, forest, baseball, residential, and parking lot.



(a) (b)

Fig. 6. Correlation visualization of the proposed MAI dataset. Here, gray lines show all connections between label pairs, the linewidths represent the correlation values between label pairs, and the categories with more and less occurrence are highlighted in different colors for demonstration, i.e., they are "Residential" and "Train Station" in MAI-AID-m and they are "Parking Lot" and "Runway" in MAI-UCM-m. (a) Correlation visualization of MAI-AID-m. (b) Correlation visualization of MAI-UCM-m.

in an intuitional way. The line connecting two labels represents the connection, and the width represents the correlation degree between two categories. Among them, for instance, "Residential" and "Parking Lot" have wide correlations with other categories, while "Runway" only has a few and the closest correlation is with "Airport."

2) Scene:

a) MAI-AID-s to MAI-AID-m adaptation: For this task, we consider 20 classes for single-label-to-multilabel DA, including airport, baseball, beach, bridge, commercial, farmland, forest, lake, parking lot, park, port, residential, river, roundabout, soccer field, stadium, storage tank, train station, works, and sea; 7050 images from the MAI-AID-s dataset are used as the source domain, while 3239 images from the MAI-AID-m dataset are used as the target domain. The number of source domain images in each class is imbalanced, ranging from 200 to 700. The input images are rescaled to 512 $\times$ 512. Examples can be found in Fig. 8.

b) MAI-UCM-s to MAI-UCM-m adaptation: For this task, both datasets have 17 classes, including airport, baseball, beach, commercial, farmland, forest, golf course, parking lot, port, residential, river, runway, sparse shrub, storage tank, tennis court, works, and sea. In total, 1700 images from MAI-UCM-s are used as the source domain and 1799 images from MAI-UCM-m are used as the target dataset. Input images are rescaled to 224 $\times$ 224. Examples are shown in Fig. 8.

3) Setup: A pretrained ResNet-101 [51] is used as the backbone of the feature generator. For different datasets, the input images are randomly cropped and resized for data augmentation. Stochastic gradient descent (SGD) is used for network optimization. The momentum is set to be 0.9, with a decay of 10^{-4}. The batch size is 4. The initial learning rate is

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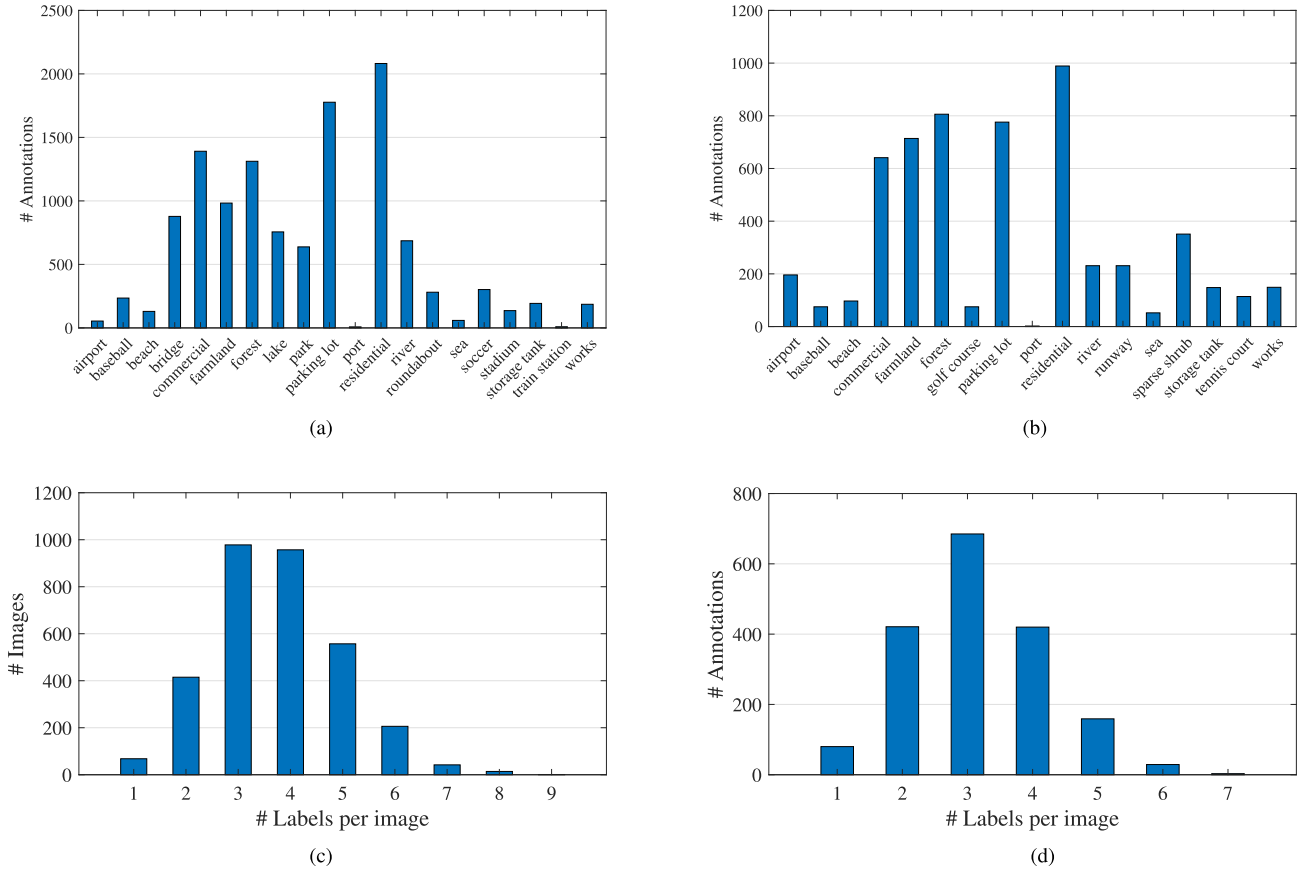


Fig. 7. Distributions of the proposed MAI dataset. (a) and (b) Categorywised annotation distributions (MAI-AID-m and MAI-UCM-m, respectively). (c) and (d) Image distributions with the number of labels in one image as the variable (MAI-AID-m and MAI-UCM-m, respectively).

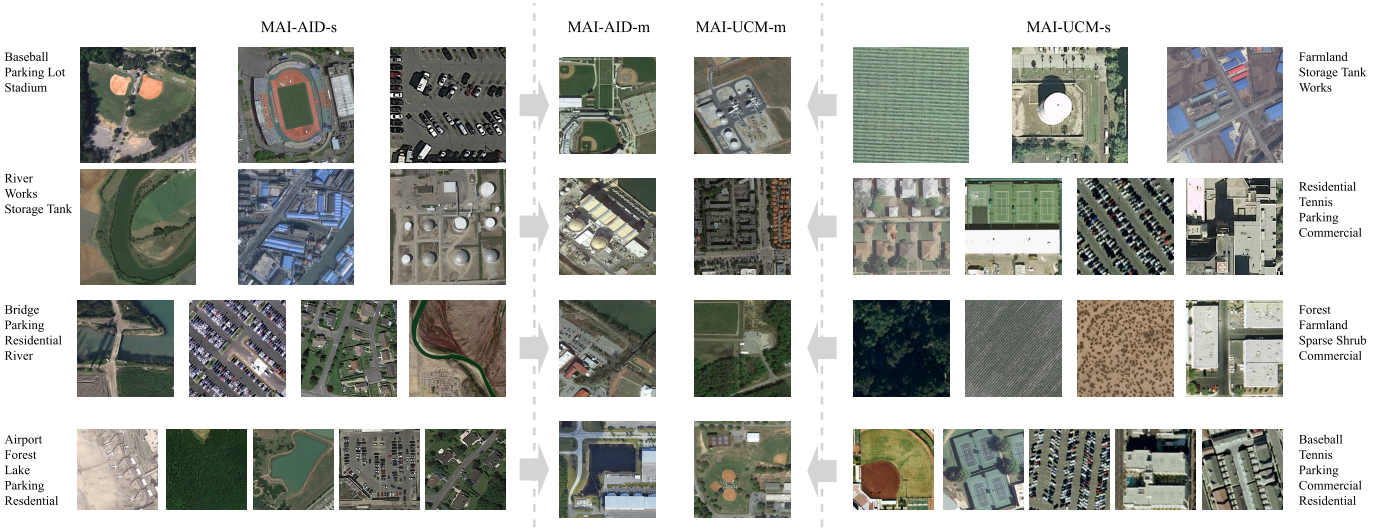


Fig. 8. Image mappings of MAI-AID-s → MAI-AID-m and MAI-UCM-s → MAI-UCM-m in the proposed DA task.

0.001 for the DWC branch and 0.01 for the LWC branch. Both learning rates decay by a factor of 10 for every 30 epochs and 200 epochs. The framework is implemented in PyTorch and trained with two 2080-TI GPUs.

4) Evaluation Metrics: In experiments, for performance evaluation, we report the average overall precision (OP), recall (OR), F1 (OF1), and F2 (OF2); and the top-3 OP, OR, OF1, and OF2 [48]. In particular, the F score (F1 when $\beta = 1$ and

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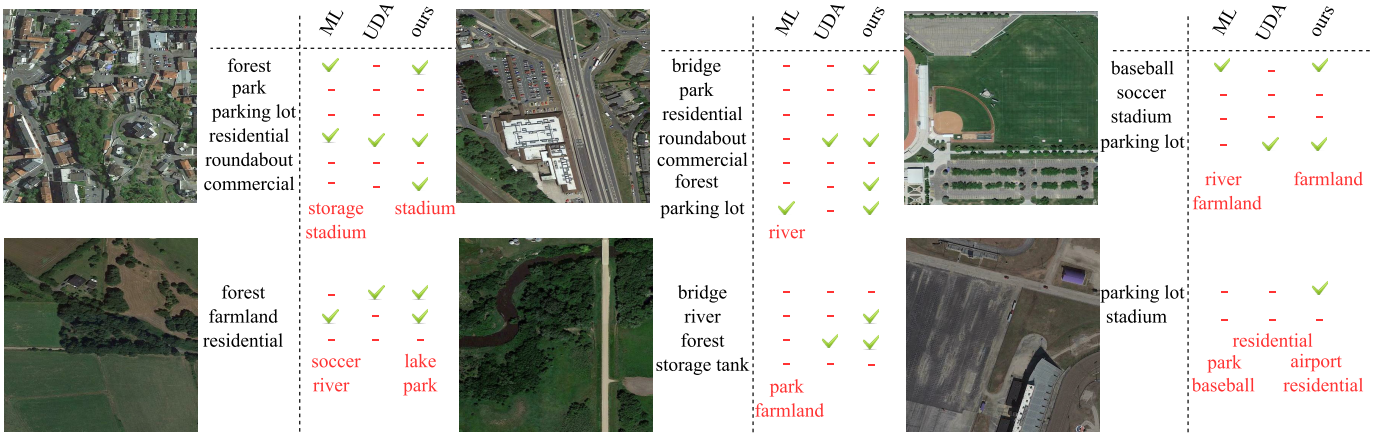


Fig. 9. MAI-AID dataset: visualization examples of the classification outputs. We use visualized results of ML-GCN [48] and MCD [44] for the results of multilabel classification (ML) and UDA, respectively.

F2 when $\beta = 2$) is calculated using

$$F_\beta = \left(1+\beta^2\right) \cdot \frac{\text{precision} \cdot \text{recall}}{\beta^2 \cdot \text{precision} + \text{recall}}. \tag{8}$$

B. Qualitative and Quantitative Comparisons

In this section, both qualitative and quantitative results of the proposed tasks and method are reported. To fully explore the capacity of our proposed network, the order and the content of the experiments are designed for specific purposes. The proposed single-label-to-multilabel DA task is novel, as there is no method reported in the literature for the proposed task yet. As the structure of the proposed framework consists of two main modules, DWC and LWC branches, we have the following:

1) comparisons with the state-of-the-art methods for multilabel classification (e.g., the best performing methods KSSNet [52] and GCN-ASL [53] and the most representative method ML-GCN [48]) to shown the effectiveness of the DWC branch;
2) comparisons with the state-of-the-art methods for DA (e.g., the best-performance UDA method HAFN/SAFN [54] and the most representative task-specific UDA method MCD [44]) to show the effectiveness of the LWC branch;
3) comparisons with the state-of-the-art methods for PLL for multilabel classification (the PRODEN [55] and DNPL methods [56]).

An intuitive comparison is shown in Fig. 9. Six images with the number of labels ranging from 2 to 7 are chosen for demonstration. The multilabel method could have multiple predictions. However, due to the domain gap between the source and the target datasets, the results are not satisfying. Though the UDA method gives a higher precision on the prediction than ML, because of its own limitation, it cannot learn the correlations between the categories of the target dataset. The proposed method is able to learn the labelwised correlation on the target domain, showing a much higher accuracy on predicting multiple labels for the proposed task.

1) Comparisons With Multilabel Classification Methods: Quantitative results are reported in Tables II and III. The results of SCIDA and SCIDA with optimized-δ are reported separately. All comparison methods are trained on the annotated single label data (MAI-AID-s/MAI-UCM-s) and tested directly on the multilabel data (MAI-AID-m/MAI-UCM-m).

For the MAI-AID-s to MAI-AID-m task, it is evident that the proposed SCIDA method provides superior classification performances under all metrics. For SCIDA with optimized-δ, the OP drops a bit because more related correlation estimation is made, while OR/OF1/OF2 improve significantly. For the MAI-UCM-s to MAI-UCM-m task, our proposed method outperforms other comparison methods under all metrics except OP. With a similar OP accuracy, the OR/OF1/OF2 are improved by about 30%. For both tasks, the optimized SCIDA consistently achieves better performances than the regular SCIDA. By introducing the LWC branch and pseudo labels, the model is able to learn category dependencies on the target datasets. Based on the performance comparisons with multilabel classification methods (without DA) in Tables II and III, we can verify that the category correlation between single-label and multilabel data learned by DA is significant.

2) Comparisons With DA Methods: Since the resolution of the MAI-UCM images is quite low, i.e., only 1/4 of that of MAI-AID images, almost all existing DA methods fail on the MAI-UCM dataset, except our proposed method with self-correction. Therefore, we only report the performance comparisons for the MAI-AID-s and MAI-AID-m task. As shown in Table II, the proposed method significantly outperforms the comparison methods under all performance metrics. In particular, the super performance of the optimized SCIDA (with opt-δ) clearly demonstrates the effectiveness of our proposed self-correction module (LWC branch). Conventional DA methods fail to capture the category dependencies on the target domain.

3) Comparison With PLL Methods: PLL is a typical weakly supervised learning problem, where each training instance consists of data and a set of candidate labels containing

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TABLE II

MAI-AID DATASET: CLASSIFICATION ACCURACY COMPARISONS WITH DIFFERENT IMAGE CLASSIFICATION METHODS. MULTILABEL METHODS: KSSNET, GCN-ASL, ML-GCN, AND DL-DECODER. DA METHODS: SAFN, HAFN, AND MCD. PARTIAL LEARNING METHODS: PRODEN AND DNPL

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
KSSNet [52]	0.2907	0.1616	0.2077	0.1774	0.3336	0.1303	0.1874	0.1484
GCN-ASL [53]	0.2026	0.1615	0.1797	0.1683	0.2742	0.0389	0.0681	0.0470
ML-GCN [48]	0.3831	0.0452	0.0809	0.0549	0.3837	0.0447	0.0801	0.0543
ML-Decoder [57]	0.2630	0.2452	0.2538	0.2486	0.2991	0.0677	0.1104	0.0801
SAFN [54]	0.4542	0.1216	0.1918	0.1425	0.4542	0.1216	0.1918	0.1425
HAFN [54]	0.4281	0,1147	0.1809	0.1344	0.4281	0.1147	0.1809	0.1344
MCD [44]	0.3327	0.0891	0.1406	0.1044	0.3327	0.0891	0.1406	0.1044
PRODEN [55]	0.1559	0.0422	0.0664	0.0494	0.1559	0.0422	0.0664	0.0494
DNPL [56]	0.1031	0.0402	0.0578	0.0458	0.1031	0.0402	0.0578	0.0458
SCIDA	**0.5432**	0.2230	0.3162	0.2528	**0.5496**	0.2196	0.3138	0.2496
SCIDA(opt-δ)	0.4474	**0.3242**	**0.3760**	**0.3431**	0.4725	**0.3185**	**0.3805**	**0.3407**

TABLE III

MAI-UCM DATASET: CLASSIFICATION ACCURACY COMPARISONS WITH DIFFERENT IMAGE CLASSIFICATION METHODS

Method	All				Top 3			
	OP	OR	OF1	OF2	OP	OR	OF1	OF2
KSSNet [52]	0.2817	0.1804	0.2199	0.1944	0.2829	0.1769	0.2177	0.1912
GCN-ASL [53]	0.1844	0.3200	0.2340	0.2790	0.1579	0.0487	0.0745	0.0565
MC-GCN [48]	**0.3585**	0.0404	0.0726	0.0491	**0.4127**	0.0356	0.0656	0.0436
ML-Decoder [57]	0.2074	0.2832	0.2394	0.2639	0.1876	0.0773	0.1095	0.0876
PRODEN [55]	0.1421	0.0560	0.0803	0.0637	0.1421	0.0560	0.0803	0.0637
DNPL [56]	0.1143	0.0478	0.0674	0.0541	0.1143	0.0478	0.0674	0.0541
SCIDA	0.3358	0.3105	0.3227	0.3153	0.3412	0.2995	0.3190	0.3070
SCIDA(opt-δ)	0.3371	**0.3219**	**0.3293**	**0.3248**	0.3380	**0.3192**	**0.3284**	**0.3228**

a unique ground-truth label. The task setting is the same as our source domain dataset. The difference is that our target domain consists of an uncertain number of labels. The candidate labels in PLL are set to be the number of the classes in the experiments. Besides, all the algorithms have the same training, testing dataset, and the same number of annotations. Tables II and III show the results on the MAI-AID and MAI-UCM datasets, respectively. Similar to DA methods, partial label methods cannot learn the label dependencies on the target domain. The proposed model surpasses the two competitors on both datasets. In particular, compared with the two PLL methods, the proposed method improves the F1 and F2 scores by more than 0.3.

C. Ablation Studies

In this section, we perform ablation studies on other parts of the framework which are not covered in the above experiments. Based on the task of MAI-AID-s to MAI-AID-m adaptation, the ablation studies are separated into the following aspects:

1) the effectiveness of GCN module;
2) the effectiveness of LWC branch;
3) the comparison of different loss functions;
4) the influence of the parameter δ;
5) stability of the proposed framework;
6) the results under the same annotation budget.

TABLE IV

ABLATION STUDY: CLASSIFICATION ACCURACY COMPARISONS OF THE PROPOSED SCIDA WITH AND WITHOUT GCN

Method	All			
	OP	OR	OF1	OF2
SCIDA(None-GCN)	0.3423	**0.3688**	0.3551	**0.3641**
SCIDA	**0.4474**	0.3242	**0.3760**	0.3431
	Top-3			
SCIDA(None-GCN)	0.3652	0.2945	0.3260	0.3060
SCIDA	**0.4725**	**0.3185**	**0.3805**	**0.3407**

1) GCN Module: In this study, we compare the results of the proposed framework SCIDA with and without GCN module to verify the importance of GCN. The results are shown in Table IV. According to the result, we could verify that, by introducing the GCN module, the performance of SCIDA is much improved.

2) With/Without LWC Branch: In this section, we generally evaluate the effectiveness of the proposed LWC branch. By deleting the LWC branch directly and only optimizing the DWC branch, we can get around 0.14 for OF1 and 0.10 for OF2. In comparison, SCIDA gets 0.38 for OF1 and 0.34 for OF2. This result verifies that the proposed LWC module is quite necessary for our framework.

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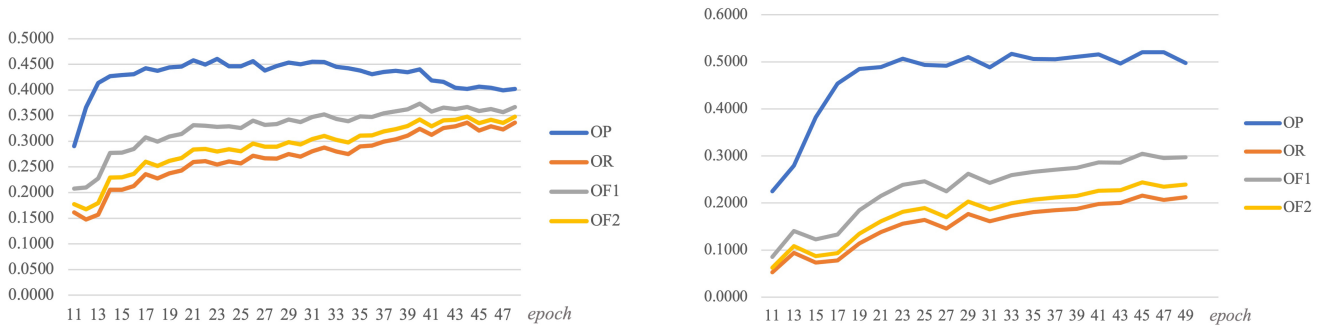


Fig. 10. Ablation study: training curves of SCIDA with different values of δ. The evaluation measures OP, OR, OF1, and OF2 of every epoch are logged until the model converges and is stable. To better show the figures, the initial ten epochs are skipped when plotting. (a) Training curve when $\delta = 0.15$. (b) Training curve when $\delta = 0.2$.

TABLE V
ABLATION STUDY: COMPARISONS WHEN USING DIFFERENT LOSS FUNCTIONS

Method	All			
	OP	OR	OF1	OF2
BCE Loss	0.1747	0.2475	0.2048	0.2285
wFL	**0.4546**	**0.2878**	**0.3524**	**0.3106**
	Top-3			
BCE Loss	0.1480	0.0764	0.1008	0.0846
wFL	**0.4654**	**0.2806**	**0.3501**	**0.3048**

3) BCE Loss Versus wFL: In this section, we evaluate different loss functions for step-1 training. In particular, we investigate two loss functions, including the widely used BCE loss and the proposed wFL. To make a fair comparison, hyperparameters under these two loss functions are tuned specifically to achieve the best performance. The results of the two loss functions are both selected when the model is converged and stable. Table V shows the results using different loss functions on the MAI-AID-s and MAI-AID-m task. Because of the existence of both severe inner-class and interclass data imbalance, the proposed weighted loss function is able to mitigate the effect of the imbalance to some extent and achieve a higher accuracy.

4) Different Values of δ for LWC Branch: To explore the effects of δ on classification performance, we consider different values of δ, ranging from 10% to 25%, as shown in Fig. 11. We note that, when δ is set as 20%, the performance is the best. If δ is too small, the correlation learning in the GNN will be slow and the training of the branch will be insufficient; while when δ is larger than 20%, redundant and useless connections will seriously affect the LWC branch, resulting in a worse overall performance. Therefore, we empirically set δ as 20%.

5) Model Stability: The proposed framework SCIDA consists of several components. The stability of the framework is quite critical. In fact, the two-stage training manner introduced in Section IV could ensure the stability of the model to a great extent. To verify that, the training curve of SCIDA under different δ values is plotted, as shown in Fig. 10. The proposed method is quite stable under different conditions.

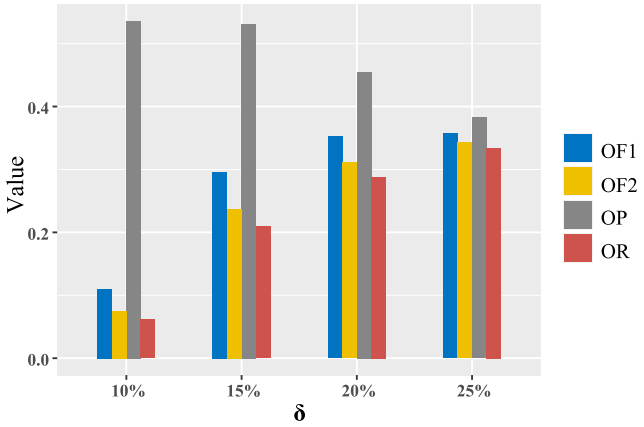


Fig. 11. Ablation study: performance evaluation when using different δ's.

6) Same Budget of Annotations: In this experiment, we want to demonstrate that with the same number of annotations, when compared with the method directly trained on the multilabel dataset, the proposed transfer method can yield comparable results. First, the proposed SCIDA uses 5000 images from the AID dataset for training. In comparison, we train the state-of-the-art multilabel classification method ML-GCN using the target MAI-AID images directly. As the average number of annotations per image for MAI-AID is 3.4, 1500 images (with $3.4 \times 1500 \approx 5000$ labels) are chosen randomly for training and the remaining are used for testing. With the same number of annotations as described above, compared with the 0.4355 (OF1) and 0.3820 (OF2) of ML-GCN which is trained directly on the target data, SCIDA could achieve comparable results as 0.3673(OF1) and 0.3493 (OF2). It is worth emphasizing that the proposed SCIDA only uses publicly available annotated single-label images. The results suggest that for model training, prior knowledge from publicly available single-label images can be an efficient alternative to manual annotations of multilabel images.

VI. CONCLUSION

In this article, we propose a novel framework for single-label-to-multilabel aerial scene adaptation task. The proposed SCIDA model integrates self-correction to DA and achieves

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superior performance. This model can be applied to large-scale, unlabeled, and unconstrained aerial images. The model is trained in a two-stage manner. Our reported multilabel classification results in the target domain demonstrate the effectiveness of the proposed model. A new MAI dataset is collected and used for experiments. For future work, we will extend the proposed SCIDA model to more challenging image types and applications.

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